EMPLOYEE ATTRITION PREDICTION USING MACHINE LEARNING

Data Science Research Capstone & Whitepaper

Research focus: Evaluating a supervised-learning baseline for employee attrition classification using the IBM HR Analytics Employee Attrition & Performance dataset.

Prepared as: Academic / portfolio research prototype

Model: Logistic Regression with preprocessing pipeline

Previously recorded baseline: 86.05% accuracy; attrition recall 0.34

Important limitation: This report describes an educational prototype. It is not validated for real-world HR decisions.

Abstract

Employee attrition creates recruitment, onboarding, continuity, and knowledge-retention costs. This capstone examines whether a supervised machine-learning model can classify employee attrition from structured HR attributes. The study uses the public IBM HR Analytics Employee Attrition & Performance benchmark, commonly distributed with 1,470 employee records. The workflow covers data validation, removal of identifier or constant fields, stratified train-test splitting, preprocessing of numeric and categorical variables, Logistic Regression training, evaluation, and model export. A previously recorded baseline run reported 86.05% accuracy, 0.62 precision, 0.34 recall, and 0.44 F1-score for the attrition class. These results show why accuracy alone is insufficient: the baseline correctly classifies many non-attrition cases but misses a substantial share of attrition cases. The project therefore treats the model as an exploratory decision-support prototype and recommends class-sensitive evaluation, calibration, fairness assessment, and human-centered interventions before any operational deployment.

1. Introduction

Employee turnover is a meaningful organizational challenge because it can disrupt teams, increase replacement costs, and lead to loss of institutional knowledge. Data science may help organizations understand patterns associated with attrition, but prediction is not the same as explanation or causation. A model may identify statistical associations in historical data without establishing why an employee leaves.

This project provides a reproducible baseline workflow suitable for a student portfolio. It is designed to demonstrate a complete data science lifecycle rather than claim a production-ready HR solution.

2. Research Question and Objectives

Research question: How effectively can a supervised machine-learning baseline identify employee attrition from available HR attributes, and what are the practical limitations of using such predictions?

- Prepare and validate the HR dataset for modeling.
- Train a reproducible Logistic Regression baseline with a held-out test set.
- Evaluate the model using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC.
- Document the limitations and responsible-use requirements of employee attrition prediction.
- Package the workflow as a repository that another student can reproduce.

3. Literature Background

Research on employee turnover emphasizes that leaving decisions are influenced by multiple individual, organizational, and contextual factors. Hom, Lee, Shaw, and Hausknecht (2017) review the development of turnover theory and highlight the variety of processes that lead employees to leave. Allen, Bryant, and Vardaman (2010) discuss evidence-based approaches to retention, emphasizing that organizations should diagnose specific turnover problems rather than apply generic interventions. Hausknecht and Trevor (2011) review turnover and retention research at the unit level, showing that turnover outcomes can have different implications depending on context.

These findings motivate caution in translating predictive scores into actions. Historical HR records may encode organizational practices and inequities. Predictive performance alone does not demonstrate that a variable causes attrition, nor does it prove that an intervention will improve retention.

4. Dataset and Data Understanding

The benchmark commonly known as IBM HR Analytics Employee Attrition & Performance contains 1,470 rows and a target column named *Attrition*, with values Yes and No. The previously recorded distribution was 1,233 No cases and 237 Yes cases, meaning the positive class represents approximately 16.1% of the dataset. This imbalance makes it possible to obtain apparently strong accuracy while still missing many employees who leave.

Available fields include demographic and work attributes such as age, department, job role, business travel, overtime, job satisfaction, years at the company, total working years, monthly income, and other HR measures. The dataset is a benchmark and should not be assumed to represent the population, policies, or employee experiences of any particular employer.

5. Methodology

5.1 Data validation and cleaning

The training script checks that the target column exists, removes duplicate rows, normalizes target labels, and retains only Yes/No target values. Identifier-like and constant fields such as EmployeeNumber, EmployeeCount, Over18, and StandardHours are removed when present. The target is encoded as 0 for No and 1 for Yes.

5.2 Train-test split

The baseline uses an 80/20 stratified split with random_state=42. Stratification preserves approximately the same target-class proportions in the train and test sets. The previously recorded split contained 1,176 training rows and 294 test rows.

5.3 Preprocessing and model

Numeric variables are imputed using the median and standardized. Categorical variables are imputed using the most frequent value and one-hot encoded, with unknown categories ignored at inference time. These transformations and Logistic Regression are placed in a single scikit-learn Pipeline to reduce leakage risk and keep training and inference transformations consistent. The model uses a maximum of 3,000 iterations.

5.4 Evaluation

The project reports accuracy, precision, recall, F1-score, a confusion matrix, a classification report, and ROC-AUC. The positive class is Attrition=Yes. Because attrition is the minority class, recall and precision for this class are particularly important. The 0.50 classification threshold is a default, not necessarily the best business threshold.

6. Results

The earlier baseline run recorded the following metrics. They are included as the project's existing baseline; the supplied training script recalculates metrics on the dataset provided by the user, and exact results may vary with preprocessing and software versions.

Metric	Previously recorded value
Accuracy	86.05%
Attrition precision	0.62
Attrition recall	0.34
Attrition F1-score	0.44
Test rows	294

Interpretation: 86.05% accuracy suggests the baseline classifies the overall test set reasonably well, but attrition recall of 0.34 means it identifies only about one-third of the actual attrition cases at the chosen threshold. Precision of 0.62 means that among cases predicted as attrition, roughly 62% were positive in that run. The F1-score of 0.44 reflects the trade-off between precision and recall. For a retention-oriented use case, false negatives may be costly, but lowering the threshold will usually increase false positives. The appropriate operating point depends on the intervention, available resources, and employee experience.

7. Business Implications

If validated, aggregate attrition risk patterns could help an organization prioritize listening sessions, review workload and overtime patterns, assess career-development opportunities, or examine manager support and work-life

balance. The model should never be used to label an individual as disloyal, automatically deny opportunities, or make punitive decisions. Interventions should be supportive, voluntary where appropriate, and evaluated for effectiveness.

- Use outputs to generate questions for organizational learning, not to make decisions about individuals.
- Prefer aggregate reporting with privacy-preserving minimum group sizes.
- Evaluate whether model errors differ across relevant groups and job families.
- Measure whether any intervention improves employee outcomes rather than optimizing prediction alone.

8. Limitations, Ethics, and Validity

The dataset is a limited benchmark and may be synthetic or anonymized; its feature definitions, collection context, and time period may not match a real employer. The sample may contain selection bias, label imbalance, and historical patterns that do not generalize. The recorded metrics are a single hold-out result, not confidence intervals or external validation. The reported minority-class recall is low. Logistic Regression may underfit non-linear relationships, while more complex models may introduce interpretability and fairness challenges.

A responsible evaluation should include calibration, threshold analysis, subgroup performance, data provenance review, privacy assessment, and external validation. Protected or sensitive attributes require particular care. In many employment contexts, automated employment decision systems may be subject to legal and regulatory requirements; obtain qualified legal, privacy, and HR review before any real-world use.

9. Future Work

- Compare Logistic Regression with a class-weighted baseline, Random Forest, and gradient-boosted models using the same held-out split or cross-validation.
- Use precision-recall curves and threshold analysis to select an operating point aligned to a clearly defined intervention.
- Report confidence intervals and cross-validation variability.
- Assess calibration and subgroup error rates, with appropriate privacy safeguards.
- Perform feature ablation and permutation importance cautiously; treat them as associations, not causal explanations.
- Test on newer, authorized, representative data before considering deployment.

10. Conclusion

This capstone delivers a complete baseline workflow for employee attrition classification and documents the limits of its predictive performance. The earlier Logistic Regression run reported 86.05% accuracy but only 0.34 recall for Attrition=Yes, demonstrating that overall accuracy can hide weak detection of the class of interest. The strongest conclusion is not that the model can reliably identify who will leave, but that reproducible modeling and careful evaluation can reveal both useful patterns and important shortcomings. Any future application must prioritize employee well-being, fairness, privacy, transparency, and human judgment.

References

1. Hom, P. W., Lee, T. W., Shaw, J. D., & Hausknecht, J. P. (2017). One hundred years of employee turnover theory and research. *Journal of Applied Psychology, 102*(3), 530–545. https://doi.org/10.1037/apl0000103

2. Allen, D. G., Bryant, P. C., & Vardaman, J. M. (2010). Retaining talent: Replacing misconceptions with evidence-based strategies. *Academy of Management Perspectives, 24*(2), 48–64. https://doi.org/10.5465/amp.24.2.48

3. Hausknecht, J. P., & Trevor, C. O. (2011). Collective turnover at the group, unit, and organizational levels: Evidence, issues, and implications. *Journal of Management, 37*(1), 352–388. https://doi.org/10.1177/0149206310383910

4. IBM HR Analytics Employee Attrition & Performance dataset. Public benchmark commonly distributed via Kaggle; verify the dataset source and license before redistribution.

Appendix A. Reproducibility Checklist

- Place the authorized CSV at data/WA_Fn-UseC_-HR-Employee-Attrition.csv.
- Install requirements from requirements.txt.
- Run python src/train.py and retain models/metrics.json for the actual run.
- Inspect the confusion matrix and positive-class recall before interpreting results.
- Keep the dataset and saved model out of public GitHub unless sharing is permitted.